Que 1

$$\overbrace{a+b+c}^{\text{First Group}}+\underbrace{x+y+z}_{\text{Second Group}}=\overbrace{\underbrace{a+b}_{\text{Sum 1}}+c}^{\text{Total}}$$

Que 2

$$A=\begin{bmatrix} & x_1 & x_2 & x_3 \\ y_1 & a_{11} & a_{12} & a_{13} \\ y_2 & a_{21} & a_{22} & a_{23} \\ y_3 & a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Que 3

$$f(x)=\begin{cases} x^2+2x+1, & x<0, \\ \sqrt{x+1}, & 0\le x<4, \\ \dfrac{x^2-1}{x-1}, & x\ge 4. \end{cases}$$

Que 4

$$\begin{bmatrix} \overbrace{a_{11} & a_{12} & a_{13}}^{\text{Column Group 1}} & \overbrace{a_{14} & a_{15}}^{\text{Column Group 2}} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \end{bmatrix}$$

Que 5

$$F(x)=\underbrace{\overbrace{\sum_{i=1}^{n}\left(a_i x_i^2+b_i x_i+c_i\right)}^{\text{Quadratic Contribution}}+\overbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}}^{\text{Matrix Contribution}} A \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{\text{Total Energy}}$$

where

$$A=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Que 6

Roll No.	Name	Department	Semester	CGPA
101	Amit	CSE	2	8.45
102	Neha	CSE	2	9.12
103	Rahul	ECE	4	8.76
104	Priya	ECE	4	9.01

Que 7

Academic Performance		
Student	**Midterm**	**Final**
A	78	89
B	85	92
C	91	94

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Department	Course	Credits	Grade
CSE	Data Structures	4	A
	Machine Learning	4	A+
	Operating Systems	3	B+
ECE	Digital Electronics	4	A
	Signals	3	B+

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Model	Parameters	FLOPs	Accuracy	Energy (kWh)
ResNet18	11.7M	1.8G	94.2%	0.00093
MobileNetV2	3.5M	0.3G	92.8%	0.00061
EfficientNet-B0	5.3M	0.39G	93.5%	0.00072

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Student	Mathematics	Physics	Computer Science
Amit	92	81	95
Neha	72	88	91
Rahul	68	74	79
Priya	96	94	98

Que 11

Technique	**Description**	**Advantages**	**Limitations**
Pruning	Removes unnecessary parameters from a neural network.	Reduces model size and computation.	May reduce accuracy.
Quantization	Represents weights using lower precision.	Reduces memory and energy consumption.	Can introduce numerical error.
Knowledge Distillation	Transfers knowledge from a large teacher model to a smaller student model.	Produces compact models.	Requires training a teacher model.

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Table 1: Comparison of model complexity, accuracy and energy usage.

Model	Parameters	FLOPs	Accuracy	Energy
ResNet18	11.7M	1.82G	94.21	0.00093
MobileNetV2	3.5M	0.30G	92.84	0.00061
EfficientNet-B0	5.3M	0.39G	93.52	0.00072

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Model	**Architecture**	**Performance**	
CNN	Conv → ReLU → Pool	**Metric**	**Value**
		Accuracy	94%
		F1 Score	93%
ResNet	Conv → BN → ReLU	**Metric**	**Value**
		Accuracy	96%
		F1 Score	95%
ViT	Attention → MLP	**Metric**	**Value**
		Accuracy	97%
		F1 Score	96%

Que 14

Model	**Loss Function**	**Parameters**	**Complexity**
Linear	$L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	10K	$O(n)$
CNN	$L = -\sum y \log(\hat{y})$	1.2M	$O(n^2)$
Transformer	$L = -\sum_{i} y_i \log p_i$	110M	$O(n^2 d)$